Program Structure for Third Year

W.E.F. A.Y. 2027-28

Semester V

Course Code	Course Name	Category	Teaching Scheme (Contact Hours)			Credits Assigned			
			Theory	Pract.	Tut.	Theory	Pract.	Tut.	Total
AIDSC501	Artificial Intelligence	PCC	3	-	-	3	-	-	3
AIDSC502	Data Engineering	PCC	3	-	-	3	-	-	3
AIDSC503	Data warehousing & Mining	PCC	3	-	-	3	-	-	3
AIDSPE501X	Program Elective-I	PEC	3	-	-	3	-	-	3
MDMC50X2	Multi-Disciplinary course-II	MDM	3	-	-	3	-	-	3
AIDSL501	Artificial Intelligence Lab	PCC	-	2	-	-	1	-	1
AIDSL502	Data Engineering Lab	PCC	-	2	-	-	1	-	1
AIDSL503	Data warehousing & Mining Lab	PCC	-	2	-	-	1	-	1
AIDSL504	Interpersonal & Career Skills	HSSM (AEC)	-	2*+2	-	-	2	-	2
MDML50X1	Multi-Disciplinary Minor course-II Lab/Tut	MDM	-	2&	-	-	1&	-	1
AIDSM501	Mini Project 2	Project	--	2#	--	--	1	--	1
Total			15	14	--	15	7	-	22

Examination Scheme - AIDS Semester-V

Course Code	Course Name	Examination Scheme						
		Theory				CIAP	ESEP	Total
		Internal Assessment		ESE$	Exam Duration (Hrs.)			
		ISE	MSE					
AIDSC501	Artificial Intelligence	20	20	60	3	-	-	100
AIDSC502	Data Engineering	20	20	60	3	-	-	100
AIDSC503	Data warehousing & Mining	20	20	60	3	-	-	100
AIDSPE501X	Program Elective-I	20	20	60	3	-	-	100
MDMC50X2	Multi-Disciplinary course-II	20	20	60	3	-	-	100
AIDSL501	Artificial Intelligence Lab	-	-	-	-	25	25	50
AIDSL502	Data Engineering Lab	-	-	-	-	25	-	25
AIDSL503	Data warehousing & Mining Lab	-	-	-	-	25	25	50
AIDSL504	Interpersonal and Career Skills	-	-	-	-	50	-	50
MDML50X1	Multi-Disciplinary Minor course-II Lab/Tut	-	-	-	-	25	-	25
AIDSM501	Mini Project -2	-	-	-	-	25	25	50
Total		100	100	300	-	175	75	750

* Two hours of practical class to be conducted for full class as demo/ discussion.

Indicates workload of a learner (Not faculty) for Mini Project 1A. Faculty Load: ½ hour per week per four groups.

$ ESE of duration 3 hours are of 80 marks and scaled to 60. ESE duration of 2 hours are of 60 marks and scaled to 45.

& Tuts only for Management MDM (MDMC5062)

Course Code	Course Name	Teaching Scheme (Hrs.)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical	Tutorial	Total
AIDSC 502	Data Engineering	3	-	-	3	-		3

Course Code	Course Name	Examination Scheme					
		Theory Marks			CIAP	ESEP	Total
		Course Assessment		ESE[s]			
		ISE	MSE				
AIDSC 502	Data Engineering	20	20	60	-	-	100

Pre-requisite:

1. Database Management Systems
2. Python Programming
3. Operating System

Program Outcomes Addressed

1. PO1 – Engineering Knowledge
2. PO2 – Problem Analysis
3. PO3 – Design/Development of Solutions
4. PO5 – Modern Tool Usage
5. PO6 – Engineer & Society
6. PO7 – Environment & Sustainability
7. PO12 – Life-long Learning

Course Objectives:

1. To provide foundational knowledge of data engineering concepts, lifecycle, responsibilities, and modern data stack components.
2. To develop an understanding of data generation mechanisms in operational systems, storage architectures, and principles of data reliability, consistency, and quality.
3. To enable learners to design and implement data ingestion workflows, understand streaming/batch ingestion, and apply basic data lineage analysis.
4. To impart skills in data orchestration, modeling approaches, transformation techniques, and their role in analytical and operational ecosystems.
5. To introduce distributed data processing frameworks such as Hadoop and Spark, and provide hands-on experience with parallel computation concepts.
6. To familiarize students with data serving layers, data security fundamentals, and emerging trends, including real-time pipelines, data mesh, and ML-integrated data architectures.

Course Outcomes: Learners will be able to

1. Explain the concepts, scope, and lifecycle of data engineering, including data formats, quality dimensions, and engineering roles.

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2. Describe and differentiate various data generation mechanisms, storage systems, consistency models, and cloud-native architectures for data management.
3. Design and implement basic data ingestion workflows for batch and streaming systems, apply serialization techniques, and analyze data lineage within data pipelines.
4. Construct data models such as normalized schemas, star/snowflake schemas, and implement orchestration workflows using tools such as Airflow.
5. Apply distributed data processing principles using Hadoop and Spark, perform transformations, optimize execution workflows, and evaluate system performance.
6. Develop a conceptual data serving layer, incorporate security and governance controls, and examine future trends in real-time data engineering and ML-driven architectures.

Module No.	Unit No.	Topics	Hrs.	Mapped to Course Outcome
1.0	**01**	**Foundations of Data Engineering**	**05**	
	1.1	Understanding Data Engineering: Definition, Architecture, Role, & Scope. Data Engineering Lifecycle: Generation-Storage- Ingestion Transformation- Serving, OLTP vs OLAP		**CO1**
	1.2	Data types & formats, Data Quality Fundamentals: accuracy, completeness, consistency, lineage, validation rules.		
	1.3	Data Engineering Responsibilities & Skills: • Technical responsibilities: pipelines, systems, integrations • Business responsibilities: data quality, governance, collaboration • Org-level setup: internal vs external data engineers		
		Self-learning Topics: Data maturity levels • Emerging tools and modern data stack snapshot • High-level case studies (e.g., Netflix, Uber, Swiggy) • Hands-on exploration of datasets • Converting between common data formats • Simple modelling of a business dataset into star schema		
2.0	**02**	**Data Generation & Storage Architectures.**	**07**	
	2.1	Change Data Capture (CDC): • Database Logs, CRUD, insert-only models. • Real-time event generation (Kafka-style messaging)		**CO2**
	2.2	Data Generation in Source System: Operational DBs (OLTP), API systems, log-based systems, Batch vs Streaming data generation.		
	2.3	Storage Fundamentals: File, block, and object storage, Serialization, compression, and caching basics, GCP Big Query, Cloud Storage. Data Storage Systems: Distributed storage concepts, Eventual vs strong consistency, Indexing, partitioning, clustering fundamentals		
	2.4	Data Warehouses, Lakes & Lakehouses: When to use what, Cloud-native storage architecture, Delta Lake architecture, Data Catalogs, schema management, Time travel, Table optimization (Z-order, vacuum)		

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		Self-learning Topics. Reading server logs, Sample API pulls, Modern File Formats (Parquet, Avro, ORC), Introduction to Delta Lake / Iceberg (Lakehouse Table Formats).		
3.0	**03**	**Data Ingestion and Lineage Analysis.**	**06**	
	3.1	Ingestion Fundamentals: Bounded vs unbounded data, Stream ingestion (Kafka, AWS Kinesis conceptually)		**CO3**
	3.2	Methods of Ingestion, Direct DB ingestion, API ingestion, Message queue & event streaming, File-based ingestion (CSV/JSON/Parquet)		
	3.3	Engineering Concerns: Serialization/deserialization, Throughput, latency, durability, DLQs (Dead Letter Queues), Ordering and reply.		
	3.4	Introduction to Data Lineage, Building a Data Flow, Edge Intelligence in Data Flow, Data Lineage — Benefits and Importance, Error handling strategies		
		Self-learning Topics: Batch ingestion (ELT/ETL), building a small batch pipeline using Python, a Simple streaming pipeline using Kafka / Redpanda / MQTT, and applying data cleaning rules on real datasets		
4.0	**04**	**Data Orchestration, Modelling and Transformation**	**07**	
	4.1	Data pipeline orchestration: need and principles, Apache Airflow fundamentals: DAGs, operators, tasks, scheduling, Monitoring & logging for data systems		**CO4**
	4.2	Queries & Optimization: Query life cycle, Cost-based optimization, Practical indexing, partitioning, clustering		
	4.3	Data Modeling: Conceptual, logical, physical models, Normalization & denormalization, Data Vault concepts (light intro)		
	4.4	Transformations (Batch & Streaming): ELT workflows, Materialized views, Federation & query virtualization, Streaming transformations (windowing, late events)		
		Self-learning Topics: **Introduction to Tableau:** Creating dashboards, storyboards, interactive filters, and connecting to different data sources.		
5.0	**05**	**Distributed Data Processing**	**06**	
	5.1	Distributed computing: fundamentals and needs Characterization of Distributed Systems: Types of distributed systems (Difference between tightly coupled and loosely coupled multiprocessor systems) Distributed Computing System Models. Distributed Share Memory.		**CO5**
	5.2	Distributed File Systems: Physical Organization of Apache Spark architecture: RDDs, Data Frames, DAG scheduler, Spark transformations, and actions.		
	5.3	Cluster modes: local, standalone, Kubernetes Performance tuning basics: caching, partitioning, shuffle optimization		

		Self-learning Topics: Fault Tolerance & Data Locality in Distributed Processing, Modern Data Storage Formats & Lakehouse Basics (Parquet, ORC, Delta Lake)		
6.0	**06**	**Data Serving, Security & Future Trends**	**08**	
	6.1	**Serving Data for Analytics & ML**: Business vs operational vs embedded analytics, Data products concept, Semantic layers & metrics layers, Reverse ETL (operationalizing data)		**CO6**
	6.2	**Security & Privacy in Data Engineering**: Principle of least privilege, Encryption basics, Logging, monitoring, audits, Shared responsibility model in cloud		
	6.3	**Future of Data Engineering**: Shift towards real-time pipelines, Fusion of data + applications, Data mesh concepts, Rise of ML-integrated data systems, Live data stack and streaming-first architectures		
		Self-learning Topics: Modern OLAP Engines (DuckDB, ClickHouse, Druid, BigQuery), Data Quality Frameworks (Great Expectations, Soda, Monte Carlo)		
		Total	**39**	

Textbooks:

1. J. Reis and M. Housley, *Fundamentals of Data Engineering*. Sebastopol, CA: O'Reilly Media, 2022.
2. T. Akidau and S. Chernyak, *Streaming Systems: The What, Where, When, and How of Large-Scale Data Processing*. Sebastopol, CA: O'Reilly Media, 2018.
3. A. S. Tanenbaum and M. van Steen, *Distributed Systems: Principles and Paradigms*, 2nd ed. Upper Saddle River, NJ: Prentice Hall/Pearson, 2007.
4. H. Garcia-Molina, J. D. Ullman, and J. Widom, *Database Systems: The Complete Book*, 2nd ed. Upper Saddle River, NJ: Pearson Education, 2008.
5. R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th ed. Boston, MA: Pearson, 2017.

Reference books:

1. J. Leskovec, A. Rajaraman, and J. D. Ullman, *Mining of Massive Datasets*, 3rd ed. Cambridge: Cambridge University Press, 2020.
2. B. Antony, *Professional Hadoop*. Indianapolis, IN: Wrox/Wiley, 2016.
3. A. P. More and P. Shah, *Apache Kafka Essentials*. Birmingham, UK: Packt Publishing, 2015.
4. M. Islam et al., *Kafka Streams Real-Time Processing*. Birmingham, UK: Packt Publishing, 2018.
5. P. Ponniah, *Data Warehousing Fundamentals for IT Professionals*, 2nd ed. Hoboken, NJ: Wiley, 2010.

Online References:

1. **Google Cloud – Data Engineering Learning Path**
   https://cloud.google.com/learn/training/data-engineering-and-analytics
2. AWS Big Data & Data Engineering Documentation (Kinesis, EMR, Glue, Redshift, S3)
   https://docs.aws.amazon.com/whitepapers/latest/big-data-analytics-options/what-is.html
3. Databricks Academy – Apache Spark Learning Resources
   https://www.databricks.com/learn/training
4. Snowflake University – Data Warehousing & ELT Workflows
   https://learn.snowflake.com/en/
5. Confluent Developer – Kafka Learning Portal & Certification Material
   https://developer.confluent.io/learn-kafka/
6. NPTEL – Database Management Systems (IIT Madras)
   https://nptel.ac.in/courses/106106093
7. NPTEL – Introduction to Data Analytics (IISc Bangalore)
   https://nptel.ac.in/courses/106108171

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Course Assessment:

ISE:

- To be conducted in any of these forms - Assignment/ Quiz/ Presentation/ Class Test/ Case study etc. of 20 marks.
- ISE 20 marks = 05 marks for attendance + 15 marks for activities.

MSE:

- To be conducted as written examination for 20 marks (on 40% - 50% syllabus)

End Semester Examination:

$ ESE of duration 03 hours are of 80 marks and scaled to 60.

2. Question paper will comprise of 03 questions.
3. Question 1(20 marks) : - Solve any 04 out of 06. All questions carry 05 marks each.
4. Question 2(40 marks) : - Solve any 04 out of 06. All questions carry 10 marks each.
5. Question 3(20 marks) :- Solve any 04 out of 06. All questions carry 05 marks each.
6. All COs should be mapped as per the weightage in the syllabus.

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SIES Graduate School of Technology
Department of Artificial Intelligence(AI) and Data Science
Bachelor of Engineering

Course Code	Course Name	Teaching Scheme (Hrs.)			Credits Assigned			
		Theory	Practical	Tutorial	Theory	Practical	Tutorial	Total
AIDSL502	Data Engineering Lab	--	02	--	--	01	-	01

Course Code	Course Name	Examination Scheme					
		Theory Marks			CIAP	ESEP	Total
		Course Assessment		ESE			
		ISE	MSE				
AIDSL502	Data Engineering Lab	--	--	--	25		25

Pre- requisite:

1. Basic Python programming (pandas, numpy, data handling)
2. Fundamentals of databases (ER modeling, SQL basics)
3. Understanding of data formats (CSV, JSON, structured/semi-structured data)

Program Outcomes Addressed

1. PO1 – Engineering Knowledge
2. PO2 – Problem Analysis
3. PO3 – Design/Development of Solutions
4. PO4 – Conduct Investigations of Complex Problems
5. PO5 – Engineering Tool Usage
6. PO9 – Communication
7. PO10 – Project Management and Finance
8. PO11 – Life-Long Learning

Lab Outcomes:

1. Perform data exploration, cleaning, modeling, and interpretation across structured and semi-structured datasets.
2. Design and implement data models using relational and NoSQL paradigms, applying normalization, schema design, and flexible document modeling.
3. Build ETL and API-driven ingestion pipelines that extract, transform, validate, and load data into analytical systems.
4. Construct OLAP models such as star schemas, cubes, and multidimensional aggregations, and analyze results to generate business insights.
5. Apply modern tools for data warehousing, dashboards, workflow orchestration, and distributed-style processing to automate pipelines and interpret results.
6. Work collaboratively to build and present an end-to-end mini data platform integrating ingestion, storage, validation, transformation, and analytics.

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Suggested List of Experiments: Students are required to complete at least 12 experiments.		
Sr. No.	**Title of Experiments**	**LO**
A: Foundation — Data Exploration, Cleaning & Modeling		
1	Exploratory Data Analysis & Data Cleaning (Python) • Load CSV dataset • Handle missing values, outliers, and inconsistent types • Generate summary statistics & visualizations Outcome: Understanding data quality and profiling.	**LO1**
2	Relational Schema Design & SQL Normalization • Convert problem statement → ER Diagram • Normalize to 3NF • Implement schema in PostgreSQL • Insertple data, run analytical SQL • Window functions (dense rank, lag/lead) • Query optimization (indexes, partitions, explain plans) • OLAP analytical SQL patterns Outcome: Modeling for structured systems.	**LO2**
3	Mini ETL: CSV → Cleaned Table in PostgreSQL • Extract CSV • Transform (clean, standardize, derive fields) • Load into relational DB • Automate with a simple CLI or script Outcome: First hands-on ETL pipeline.	**LO1**
B: Warehousing, OLAP & Business Insights		
4	To design reusable business metrics using SQL views on top of a warehouse schema and analyze decision-level insights. • Define SQL views for: • Daily Revenue • Monthly Sales Growth • Repeat Customer Percentage • Query views instead of base tables Outcome: Compare readability and reuse vs raw SQL	**LO3**
5	Data Cube & Multidimensional Aggregations • Multi-dimension GROUP BY • Cube, rollup, group sets • Compare execution plans Outcome: Practical OLAP operations.	**LO4**
6	Analytics Dashboard for Business KPIs	**LO5**

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	Tools: Python/Streamlit Public • Load warehouse tables • Build 3-5 charts answering real questions • Create a mini dashboard Outcome: BI-style storytelling with data.	
C: NoSQL and API-Driven Data Engineering		
7	NoSQL Basics: CRUD & Indexing in MongoDB • Create MongoDB database • Model flexible schemas • Insert documents • Query with filters, projections & indexes Outcome: Understanding document stores.	**LO2**
8	Implementing Delta Lake Features in a Lakehouse Environment • Create a Delta table from a sample dataset. • Insert, update, or delete records to generate multiple table versions. • Demonstrate time travel by querying older versions of the Delta table. • Test schema enforcement by attempting to write with incompatible schema. • Record and interpret system responses. Tools: Option A – Local / Python Environment: • Python • PyDeltaLake library • Pandas • Local filesystem for storing Delta tables Option B – Cloud / Databricks Community Edition: • Databricks CE notebook environment • Delta Lake runtime • DBFS (Databricks File System) • Spark SQL / PySpark	**LO5**
9	Building REST APIs for Data Capture • Using Flask or FastAPI: • Create POST/GET endpoints • Validate JSON payloads • Store into MongoDB/PostgreSQL Outcome: How Data Enters Real DE Pipelines.	**LO3**
D: Distributed Data & Pipelines		
10	Event-Driven Pipelines using RabbitMQ/Redpanda (Kafka-lite) • Create producer • Create consumer	**LO5**

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	• Consume streaming data • Store into the staging table Outcome: Foundation of streaming ingestion.	
11	Distributed Data Processing Without Spark/Hadoop Use DuckDB or Polars for large-scale local batch processing: • Load large CSV • Run parallel queries • Optimize using columnar formats Outcome: Teaches "distributed-style" parallel processing without requiring Spark.	**LO4**
E: Workflow, Automation & Final Pipelines		
12	Workflow Orchestration using Airflow or Prefect • Create simple DAG • Extract → Transform → Load • Add scheduling and logging Outcome: Automation & orchestration skills.	**LO3**
13	Data Observability and Quality Validation using Expectation Suites • Create an expectation suite for specifying null checks, value ranges, column constraints, and datatype rules. • Validate a sample dataset using the expectation suite. • Generate an HTML validation report summarizing passed and failed expectations. • Interpret the report and document insights. Outcome: Trust and governance in data.	**LO5**
14	End-to-End Batch Pipeline Project • Ingest → clean → validate → transform → load • Store final data in Parquet or PostgreSQL • Maintain logs and error files Outcome: Realistic DE batch workflow.	**LO6**
15	Application Built up: Build a Mini Data Platform Groups build a pipeline combining: • API ingestion • NoSQL/SQL modeling • Validation • ETL • Analytics dashboard Outcome: Full data engineering life cycle.	**LO6**

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Textbooks:

1. Kleppmann, M. (2017) Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems. Sebastopol: O'Reilly Media.
2. Meyer, A. (2018) The Data Engineering Cookbook.
3. Bhagwan, R., et al. (2021) Data Storage Survey: Modern Data Formats, Partitioning Strategies, and Warehouse–Lake Architectures.

Reference books:

1. McKinney, W. (2022) Python for Data Analysis. 3rd edn. Sebastopol: O'Reilly Media.
2. Densmore, J. (2021) Data Pipelines Pocket Reference: Moving and Processing Data for Analytics. Sebastopol: O'Reilly Media.
3. J. Cito, P. Leitner, and H. C. Gall(2019), Engineering of Big Data Processing Systems. Cham, Switzerland: Springer.

Online Resources:

1. AWS Data Engineering Learner's Path (Free Tier Friendly)
   https://aws.amazon.com/learning-paths/data-engineer/
2. Google Cloud Big Data & ML Fundamentals (Coursera Audit Free)
3. MongoDB University Courses
   https://learn.mongodb.com
   (Excellent for schema design, aggregation pipeline)
4. Apache Airflow Documentation https://airflow.apache.org/docs/
   (Needed for workflow orchestration.)
5. Kafka (or Redpanda) Documentation
   https://redpanda.com/documentation/
   (Redpanda is easier to run on Windows)
6. FastAPI Documentation
   https://fastapi.tiangolo.com (REST API creation.)
7. DuckDB Documentation (If using local OLAP)
   https://duckdb.org/docs/ (Lightweight, easy for partitioning, Parquet

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Software Tools

1. Python 3.x + VS Code
   For EDA, ETL, data cleaning, API development, feature engineering.
2. PostgreSQL (or MySQL)
   For relational schema design, fact/dimension tables, OLTP/OLAP queries.
3. MongoDB Community Server
   For NoSQL modeling, API storage.
4. Redpanda (Kafka-compatible lightweight broker)
   Windows-friendly event ingestion for.
5. Apache Airflow (via Docker or WSL)
   Needed only for 1 lab.
6. DuckDB (Standalone EXE)
   Zero-installation, ideal for Parquet, partitioning, OLAP-style queries.

Term Work:

- Term work should consist of at least 12 experiments.
- The final certification and acceptance of term work ensures satisfactory performance of laboratory work and minimum passing marks in term work. Term work will be assessed as Continuous Internal Assessment Practical (CIAP).
- 25 Marks (Total Marks) = 20 Marks (Experiment) + 05 Marks (Attendance)

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